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SUMMARY

Earthworm, Cockroach and Frog show characteristic features in body organization. *Lampito mauritii* (earthworm) is commonly found in Tamil Nadu, its body is covered by cuticle. It has a long and cylindrical narrow body which is bilaterally symmetrical. All segments of its body are alike except the 14 to 17 segments, which are thick and dark and glandular, forming the clitellum. This helps in the formation of cocoons. A ring of S-shaped chitinous setae is found in each segment. These setae help in locomotion. Earthworm's development is direct and no larva is formed during development.

Cockroach is a typical cosmopolitan insect and exhibits all the fundamental characteristics of class Insecta. The body of the cockroach is compressed dorso-ventrally, bilaterally symmetrical, segmented and divisible into three distinct regions – head, thorax and Abdomen. The photoreceptor organ of the cockroach consists of a pair of compound eyes with mosaic vision. Segments bear jointed appendages. There are three throacic segments of each bearing a pair of walking legs. Two pairs of wings are present, one pair each on 2nd and 3rd segment. There are ten segments in abdomen. Fertilization is internal. The development of cockroach is gradual through nymphal stages (paurometabolus).

Frogs are cold blooded vertebrates – Poikilotherms. Skin is smooth and moist, Red blood corpuscles are nucleated. Eggs are laid in water. The larvae pass through an aquatic stage before metamorphosing into adult.

Birds are warm blooded, bipedal vertebrates. They have an exoskeleton of feathers. Their forelimbs are modified into wings to aid in aerial mode of life

Glossary

After-shaft – Small tuft of soft feathers near the superior umbilicus of Quill feather

Articular membrane – The non-sclerotized, flexible membrane between the segments of arthropods, and the joints of arthropod appendages.

Bipedal – Walk or stand on two feet.

Buccal cavity – The first region of the alimentary canal, between mouth and pharynx.

Chloragogen Cells – Surrounding the alimentary canal; is attributed to excretion and regeneration.

Clitellum – A regional epidermal swelling, where gland cells secrete material to form the cocoon.

Cochlea – A coiled tube found in the inner ear, essential for hearing

Cocoon – It is a bag like structure secreted by the clitellum. Eggs and sperms are deposited into it. Fertilization and development occurs within the cocoon.

Coelom – The body cavity between the body wall and the alimentary canal.

Cutaneous Respiration – Respiration through skin

Elytra – Elytra (singular elytron) are the tough forewings of beetles and earwigs. The